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Exploring the Versatility of QAOA: A Comprehensive Review

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***Abstract*—Quantum Approximate Optimization Algorithm (QAOA) is the notable advancement in quantum computing, which allows us to solve optimization problems differently. In this paper, we provide a detailed investigation of the possible applications of QAOA accompanied by an overview of previous implementations. This is where QAOA shines and can spin out solutions where ordinary methods takes millennia to converge with acceptable results for NP-hard optimization problems that involve transfer of n qubits between a quantum state. We present a comprehensive literature review in this paper, and potential areas of application where QAOA could be successfully used such as logistics, finance and Artificial Intelligence.**

***Index Terms*—Quantum computing, Quantum Approximate Optimization Algorithm, Combinatorial optimization, Machine learning, Cryptography, NP-hard problems**

I. Introduction

The notion of quantum computing is a novel paradigm of computation that utilizes quantum mechanical principles to solve problems which exceed the potential of traditional computers. The key component of quantum computing is quantum bits or qubits that are capable of being in superposition of 0 and 1. Additionally, it can be entangled, which is defined as the blocking of quantum state on the qubits even separated by a long distance. On the other hand, the position of a classical bit upon measuring is unassured, and it will be one state only. These features allow a quantum computer to explore an enormous space of solutions in parallel and make use of quantum interference. [16]

A. Variational Quantum Eigen Solver

The Variational Quantum Eigen Solver is one of the central quantum computing algorithms that provides a very convenient way of solving quantum problems of great complexity for classical computing due to its efficiency. [26] In its core, this powerful algorithm combines classical optimization and quantum computing, employing quantum principles to navigate quantum solution spaces and classical methods to improve their outcomes and obtain relevant results.

The basic idea behind VQE is related to a variational approach to quantum computing. Instead of measuring the eigenvalues of a provided Hamiltonian, a parameterized quantum circuit (ansatz) is used to prepare trial states that can be defined as candidate solutions to the eigenvalue problem. [12] Then, in next step, the trial states are measured, and energies associated with the circuit are discovered via another quantum circuit.

The iterative process of VQE is a process of taking the parameters of the proposed quantum circuit to minimize the expected energy. In some sense, this is how other classic optimization procedures usually work. This iterative process is implemented by a series of traditional optimizations that include gradient descent or evolutionary strategies. Thus, the attempt to reach the lowest energy state of the desired Hamiltonian is improved.

B. Strong Church-Turing Thesis

The Strong Church-Turing Thesis claims that any physically possible model of computation to a Turing machine is computationally equivalent. The thesis is challenged by quantum computers, whose principles are based on insights from quantum mechanics, especially superposition and entanglement. Any computation performed by a quantum computer is done exponentially faster in comparison with classical ones. This phenomena is known as quantum speedup and is made possible by quantum parallelism and quantum entanglement. Quantum parallelism means that a system can exist in multiple states at once; entanglement is a mechanism parallelizing computation as it allows sufficiently interconnected computations. Thus, quantum computing refutes Strong Church-Turing Thesis by showing that there are systems performing computation that goes beyond the limits of classical computation. [21], [25]

C. Quantum Approximate Optimization Algorithm

The QAOA is a novel concept in the realm of quantum computing, developed to efficiently solve combinatorial optimization problems. During the emergence of optimization as an essential mathematical problem in a variety of fields, this algorithm has the potential to impact many sectors. [20] With the rise of quantum computing as a disruptive technology, challenges that have traditionally been difficult to optimize efficiently can now be addressed using the QAOA algorithm. [17] It capitalizes on the properties of quantum mechanics, such as superposition and entanglement, narrowing down the maximum to evaluate the Optimization algorithm revolves around utilizing less computational power and exploring the

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solution space more expediently than the classical optimization algorithm. [11] The algorithm achieves this through variational approaches through a series of parameterized quantum spheres. [26] Therefore, this algorithm has great promise in terms of integrating combinatorial problems into quantum states that address and refine solutions iteratively through the use of classical optimization processes. [32]

In QAOA decision variables are parameters that define rotation that are applied to qubits in quantum circuit. Angles of these rotations affect how quantum state evolves during computation thus affecting solution that will be received.

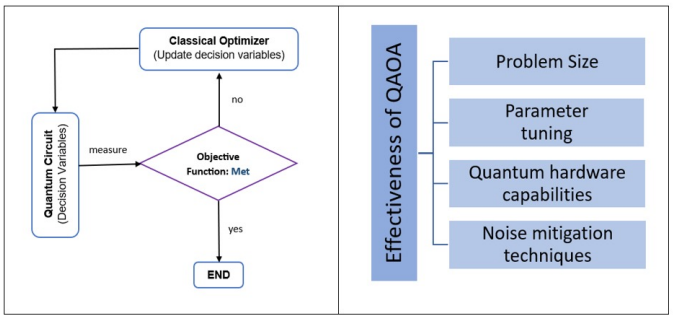

Fig. 1. QAOA working and Effectiveness of QAOA

II. Applications of QAOA

The QAOA is also a variational algorithm that is designed specifically to solve combinatorial optimization tasks. [13] The algorithm is based on preparation of quantum states that encode potential solutions to the optimization problem and optimization of the expectation value of these states with the help of parameterized quantum circuits.

A. Graph Optimization Problems

1) Max-Cut Problem: [1]An assignment for this problem is a partition of the vertices such that:

- each vertex i belongs to one of the two parts, i.e., partition;
- for every edge ij of the graph, i and j must belong to the parts, which are different in the partition.

2) Graph Coloring: Given a graph G = $\langle V, E \rangle$, three subsets of V have to be defined, such that no two adjacent vertices share some of the subsets. This state has to be reached with the minimum number of colors.

3) Graph Partitioning: Vertices of the graph G = $\langle V, E \rangle$ are in advance dividers into A and B, such that AAAA AA = or BB BB = , i.e., the classified vertices consist of one connected group.

B. Combinatorial Optimization Problems

1) Traveling Salesman Problem: [2] [3]Given N cities, the number of the shortest ways from the city J to the city K has to be found that goes on one path exactly once.

2) Quadratic Assignment Problem: [15]Dots are associated to the nearest location or a facility with the number of the facilities.

3) Maximum Clique Problem: The maximal undirected graph, where a complete graph is a subgraph

C. Resource Allocation and Scheduling

1) Job Scheduling: [5]Allocate tasks to resources over time to optimize completion time, resource utilization, or other objectives.

2) Resource Allocation: optimize the allocation of resources such as personnel, machines, or funds to maximize efficiency or minimize costs.

D. Machine Learning and Data Science

1) Feature Selection: [14], [24] Select the most relevant features from a dataset to optimize model performance.

2) Clustering and Classification: Optimize clustering algorithms or classification models to minimize error or improve accuracy. [22]

E. Energy and Resource Management

1) Energy Grid Optimization: Optimize energy distribution and resource allocation in power grids to minimize costs or maximize efficiency.

2) Resource Management: Optimize resource usage in manufacturing, transportation, or supply chain operations.

F. Network Optimization

1) Routing Optimization: Optimize routing in networks to minimize latency, congestion, or costs.

2) Network Flow Optimization: [7]Optimize flow in networks subject to capacity constraints and costs.

G. Constraint Satisfaction Problems

1) Boolean Satisfiability: Determine if a given Boolean formula can be satisfied by assigning truth values to variables.

2) Constraint Satisfaction: Find assignments to variables that satisfy a set of constraints in optimization problems.

H. Quantum Simulation

Molecular Simulation: [23] Simulate molecular structures and interactions to optimize properties for drug discovery or materials science.

I. Portfolio Optimization

Financial Portfolio Optimization: Optimize asset allocation in investment portfolios to maximize returns or minimize risk under given constraints. It uses quantum computing principles to find near-optimal solutions for decision-making problems. The target audience is anyone who has the knowledge and understanding of how investment portfolios can be allocated by assets. [19] The steps to achieve this purpose are: use QAOA and it's concave optimization algorithm and run the algorithm with different parameters. The final task is to estimate how close to optimality the algorithm can get and how much does the parameter affect it.

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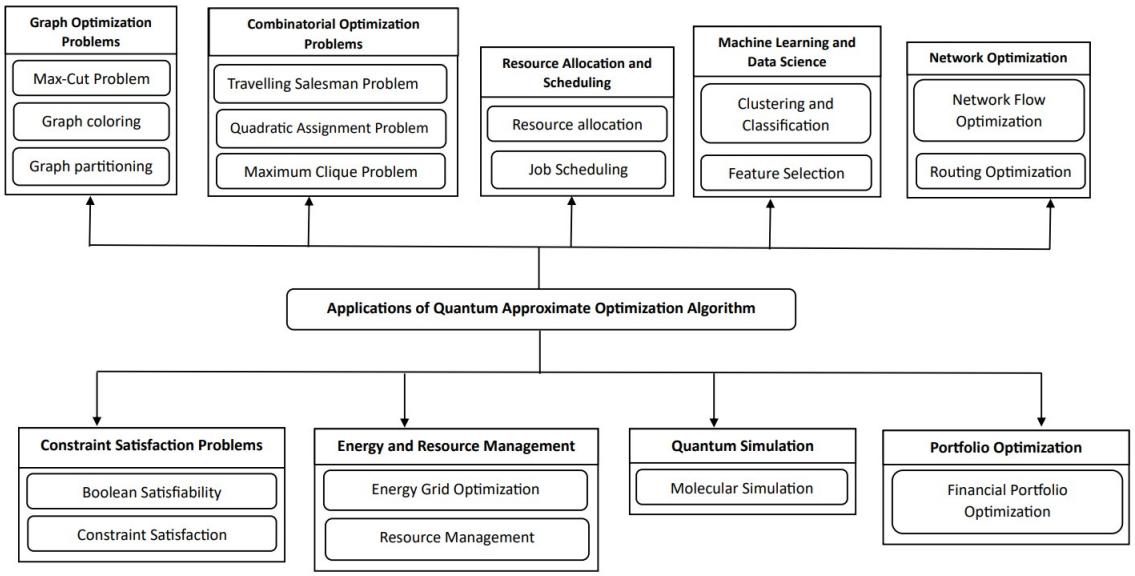

Fig. 2. Applications of QAOA

III. Applying QAOA to Optimization Problems

A. DC-QAOA

The objective of Large Graph Partitioning algorithm is to partition an undirected graph into two subsets such that parent graph's maximum size constraints are satisfied. It divides graphs by proposing separation paths and iteratively attempts to calculate the potential paths for given combinations of nodes and edges. [1] LGP algorithm determines the favorable path by ensuring that the algorithm has exactly two connected components using a depth-first search. The paths should satisfy that no crossing edges are allowed from one subset to another. Such formations are essential in keeping with measurement distribution reconstruction for quantum systems . The sub-process' measurement is used in approximating the parent system's measurements distribution. Various schemes used for distribution reconstruction include min, mul, and minXmul, which are measured using KL divergence and entropy.

The Divide-and-Conquer QAOA uses LGP and MDR in a recursive way for solving large-scale graph problems, and effectively balances the frequencies of solutions that prohibiting unavoidable exponential scaling. [21] It is concluded that LGP and MDR play important roles in both partitioning and reconstruction of a large-scale graph, when implementing the quantum algorithm on NISQ computers. [24], [25]

B. VRP

This technique focuses on constructing Variational Quantum Eigensolver for Vehicle Routing Problems using Hamiltonians and circuits with 6 and 12 qubits to exceed classical simulation input bounds. [2] However, both results with and without noise show apparent deviation from classical minimum. The key finding is that noise impacts the minimum energy costs nonlinearly with extreme noise values extending from 0.05 to 0.5. The alluded observation holds especially when considering bit-flip noise channels. Despite the alluded data, the COBYLA optimizer demonstrated its superiority compared to alternative, not least because it consistently surpasses alternative optimizers for VQE circuits across noise scenarios. [3] Thus, the alluded observation implies that the COBYLA optimizer may be harnessed to implement practical quantum algorithms designed for intermediate-scale quantum devices. The results also indicate that the alluded optimizer is not impacted by extreme noise at the first layer of the VRP circuits. As such, future work seeks to confirm these findings on more substantial VRP instances on actual devices as well as examine more complex noise models following the characteristics of real NISQ devices. [29] Also, the empirical evidence suggests that selecting the appropriate optimizers for executing noise-resilient quantum algorithms is essential and that COBYLA demonstrates moderate success even as the noise level increases.

C. VRPTW

This technique is about how to adapt Vehicle Routing Problems(VRP) with Time Windows for quantum algorithm. It presents formulations for both routes and arcs as Quadratic Unconstrained Binary Optimization. [4] Penalty terms have to be included for sequence-based formulations to deal with bi-linear equality constraints. Moreover, an Alternating Direction Method of Multipliers ADMM heuristic is proposed for VRP with Time Windows, which deals with continuous constraints using sequential convex programming. [27] Each formulation has its own benefits and drawbacks: route-based is more compact but sparse and connected poorly. The choice between QAOA and Variational Quantum Eigensolver methods is based on the availability of samples. QAOA works better with a high number of samples, whereas VQE is viable for a lower number

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of samples. The heuristic significantly converges nearly to practicable and optimal solutions. In this with use of quantum devices having noise, heuristic helps retain such qualities of solutions. Future research may include further formulation of problems with additional constraint sets and verification of quality of solutions achieved when using ADMM.

D. JSSP

This work presents the results of a study of quantum computing's potential in operational research, specifically in solving combinatorial optimization problems. It provides a brief overview of the subject and the main concepts to highlight the challenges to using quantum computations effectively. [5] Although access to NISQ devices is still limited, it is demonstrated that quantum simulations and algorithms can be developed using quantum simulators. More precisely, QAOA was adapted by the author and applied in testing on a toy JSSP instance and successfully ran on the Atos myQLM quantum simulation platform. Thus, the research showed that QAOA can be used to solve scheduling problems and obtain optimization results, solving the developed JSSP instant optimally. Also, it is shown that even hybrid classic-quantum NISQ devices successfully support similar calculations and pave their way towards bigger and more complex scheduling problems. [34] It is aimed that the children provide a basis for a deeper understanding of the principles of quantum algorithms by developers and improvement of the tools for developing and testing quantum simulations environments.

E. L-VQE

This technique evaluates the performance of the Layered-Variational Quantum Eigensolver and compares it to the QAOA. [6] In addition, we contrast the L-VQE and Variational Quantum Eigensolver with fixed ansatz. These methods are tested for their performance in solving clustering by maximizing modularity problems. The findings of the experiments conducted on different graph instances show that L-VQE exhibits better performance in terms of approximation ratio and resistance to finite-sample errors. The assessment also includes analysis of the scalability as well as the simulation of L-VQE performance using a trapped ion noisy quantum simulator. [31] Moreover, the entanglement in VQE is explored. Based on the results, it can be concluded that L-VQE can offer accurate and noise-resilient solutions in solving combinatorial optimization problems.

F. Power Network Optimization

This technique proposes a new Quadratic Unconstrained Binary Optimization formulation for the problem of power network optimization, considering the partition of graph without electricity sharing. [7] The study investigates the opportunity of applying quantum annealing in order to solve this real-life problem, proving that both pure and hybrid quantum approaches are effective. [28] The given QUBO formulation was tested using synthetic data and the German transmission power network, and it was shown that the approach outperformed the classical solvers providing the promising results. The investigation is particularly helpful in the context of the energy sector: it indicates that the quantum computing technologies can be used to resolve the challenging optimization tasks.

G. TM-QABOA

TM-QABOA presents a quantum algorithm, Quantum Approximate Bayesian Optimization Algorithm, that combines Bayesian Optimisation with a quantum circuit including phase-separating and mixer Hamilton operators which helps to increase the utilisation of optimisation. The approach includes iterating UC, UB and UG operations and then optimising rotation angles with "BO". [10] The latter is based on implementation of GPR that is metric of objective values. Another aspect, a variant of TM-QABOA, introduces a technology known as a quantum process algorithm, that outlines a uTM-QABOA algorithm which increases the robustness of optimisation because of incorporating uncertainty of measurement of basis states. The steps required include the following: preemption through fire drill, discovery, rollback, beautification, tradeoff, meetings, silos, stack ranking, values, and vision. Two versions of QAOA, TM-QABOA and uTM-QABOA, integrate Bayesian optimization to improve searching efficiency. TM-QABOA employs two mixers for exploration and exploitation, while uTM-QABOA introduces a new QM kernel based on kurtosis to dynamically control variation in objective values.

H. QAOA With Parallelizable Gates

This technique discusses a parallelizable form of QAOA using nearest-neighbor CNOT gates and single qubit rotations for solving combinatorial optimization problems. [8] The primary goal of the technique is to demonstrate computational quantum advantage with the use of the latest advances in quantum technology. In order to address a broad range of optimization problems, it encodes them into a lattice gauge model . This enables local field and interaction terms to be turned into QAOA, thus allowing their implementation with pairwise gates on a square lattice with nearest-neighbor connectivity. It was concluded that the approach would enable the solution of all-to-all connected optimization problems and could be scaled to qubits on a 2-D lattice.

I. Bayesian optimization

Bayesian optimization is a global optimization strategy tailored for efficiently finding the minimum of noisy, black-box objective functions, particularly useful when function evaluations are expensive. [9], [33] The algorithm employs a surrogate model, often a Gaussian process, to represent the objective function, updating it iteratively based on observations to form a posterior distribution. An acquisition function, derived from the posterior distribution, guides the selection of points for further evaluation. This method, requiring no prior knowledge of the objective function, is well-suited for optimizing variational circuits on Noisy Intermediate-Scale Quantum (NISQ) devices. The Gaussian process framework,with its mean and kernel functions, captures uncertainty in function evaluations, facilitating effective decision-making in optimization process.

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TABLE I
COMPARISION OF VARIOUS OPTIMIZATION PROBLEMS ON VARIOUS ASPECTS

Problem	**Cost Hamiltonian**	**Evaluation Complexity**	**Nature**	**Qubits Required**
Max-Cut	$H_{\text{Max-Cut}} = \sum_{(i,j)\in E} \frac{1 - Z_i Z_j}{2}$	$O(\lvert E\rvert)$	Quadratic (pairwise interactions)	$\lvert V\rvert$
Traveling Salesman Problem (TSP)	$H_{\text{TSP}} = \sum_{i,j} d_{ij} \left(\sum_k \frac{1-Z_{i,k}Z_{j,k+1}}{2}\right)$	$O(n^3)$	Complex (permutations)	n^2
Graph Coloring	$H_{\text{Coloring}} = A\sum_i \sum_k (1 - Z_{i,k}^2) + B\sum_{(i,j)\in E}\sum_k Z_{i,k}Z_{j,k}$	$O(\lvert E\rvert \times K)$	Unary and pairwise terms	$\lvert V\rvert \times K$
Scheduling	$H_{\text{Scheduling}} = A\sum_i \sum_t Z_{i,t} + B\sum_{\substack{(i,j)\\ \text{conflicts}}} \sum_t Z_{i,t}Z_{j,t}$	$O(M \times T)$	Unary and pairwise terms	$N \times T$
Portfolio Optimization	$H_{\text{Portfolio}} = A\sum_i x_i + B\sum_{i<j} \rho_{ij} x_i x_j$	$O(N^2)$	Linear and quadratic terms	N
Vehicle Routing Problem (VRP)	$H_{\text{VRP}} = A\sum_{i,k} Z_{i,k} + B\sum_{i\neq j} d_{ij} \left(\sum_k \frac{1-Z_{i,k}Z_{j,k+1}}{2}\right)$	$O(N^2 \times K)$	Linear and quadratic terms	$N \times K$
Maximal Independent Set	$H_{\text{MIS}} = A\sum_i Z_i + B\sum_{(i,j)\in E} Z_i Z_j$	$O(\lvert E\rvert)$	Linear and quadratic terms	$\lvert V\rvert$

TABLE II
COMPARISION OF VARIOUS QAOA TECHNIQUES

Sr. No	**Merits**	**Demerits**
DC-QAOA [1], [24], [25]	The framework addresses limitations of NISQ computers by partitioning large graphs into manageable sizes, leveraging LGP and MDR. DC-QAOA's divide-and-conquer approach allows for efficient handling of large-scale graph problems by recursively applying LGP and MDR.	MDR faces challenges in managing the exponential growth of measurement counts, which can impact computational efficiency. Balancing quantum noise and optimal solutions during reconstruction poses a challenge, potentially affecting the accuracy and efficiency of the framework.
VRP [2] [30]	The discussion thoroughly examines QAOA's performance for VRP on near-term quantum devices, considering various factors like optimizer impact and solution landscape. It offers strategies to tackle optimization challenges, including parameter initialization and exploring alternative mixer Hamiltonians, advancing QAOA optimization.	The study acknowledges QAOA's scalability limitations for larger VRP instances, hindering the demonstration of quantum advantage in real-world applications. It highlights the computational challenges in optimizing variational parameters, especially for higher circuit depths, potentially limiting QAOA's efficiency for larger problems.
VRPTW [4], [27]	The section provides various formulations of VRPTW problems suitable for quantum algorithms, accommodating different constraints and optimization objectives. The proposed ADMM-based heuristic demonstrates practical convergence to feasible and optimal solutions, offering resilience to noise inherent in quantum devices.	Reformulating VRPTW problems into QUBO form involves handling bilinear equality constraints, requiring careful consideration of penalty terms and parameter selection, potentially leading to computational complexity. The efficacy of proposed quantum algorithms heavily relies on capabilities and limitations of current quantum hardware, including qubit count, noise levels, and computational resources, which may constrain solution quality and scalability.
JSSP [5]	QAOA offers a promising approach to solving combinatorial optimization problems like JSSP on quantum computers, potentially outperforming classical algorithms in certain cases. The transformation of objectives into cost Hamiltonians provides a systematic and intuitive way to leverage quantum algorithms for optimization tasks.	Implementing QAOA on current quantum hardware may be challenging due to noise, limited qubit connectivity, and gate errors, affecting the algorithm's performance. Effectiveness of QAOA heavily relies on choice of variational parameters, which may require extensive classical optimization and may not always guarantee finding optimal solution.
L-VQE [6]	Demonstrates the superiority of L-VQE over traditional VQE and QAOA in terms of performance metrics like approximation ratio and robustness to finite sampling errors. Provides evidence of L-VQE's potential for real-world applications by conducting experiments on a trapped ion noisy quantum simulator, showcasing noise resilience and adaptability to different quantum architectures.	This primarily focuses on a specific class of combinatorial optimization problems (clustering on graphs), limiting the generalizability of findings to other problem domains. The comparison of algorithms is based on simulations rather than experiments on actual quantum hardware, which may not fully capture the complexities and limitations of real-world implementations.
Power Network Optimization [7], [28]	This presents a new technique for creating penalty terms for real-valued linear inequality constraints, improving the existing methods for handling inequality constraints and QUBO penalty functions. By focusing on a key application in the energy industry, specifically optimizing power networks, the research demonstrates the practical relevance and applicability of quantum computing in solving real-world problems.	This encounters challenges in estimating the scalability of quantum annealing with respect to problem size, particularly evident in the inability to find minor embeddings for larger problem instances. The simplifications made in the formulation, such as constant edge costs, may limit the accuracy and applicability of the approach to more complex real-world scenarios.

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Sr. No	Merits	Demerits
QAOA with Parallelizable Gates [8]	This allows for full parallelization of gates on a quantum device, enhancing efficiency and scalability in solving combinatorial optimization problems. By using LHZ mapping, the study achieves a separation of two-qubit gates from programmable local fields, offering a systematic approach to address optimization problems on quantum devices.	This technique may be restricted to specific types of optimization problems, particularly those suitable for representation on a square lattice with nearest-neighbor connectivity. Implementing this scheme require significant computational resources and may face challenges in scalability, especially for larger problem instances or more complex optimization landscapes.
Bayesian optimization [9], [33]	Bayesian optimization achieves convergence to optimal solutions with relatively few function evaluations, making it suitable for costly objective functions typical in quantum computing. The algorithm's reliance on probabilistic models allows it to handle noisy and uncertain evaluations, providing robustness in real-world optimization scenarios.	Implementing Bayesian optimization, with Gaussian processes, can be computationally intensive, especially for high-dimensional optimization problems or large-scale datasets. The performance of Bayesian optimization can be influenced by the selection of hyperparameters, including the kernel function and its parameters, requiring careful tuning and optimization, which can be challenging in practice.
TM-QABOA [10]	BO enables global optimization without requiring explicit knowledge of the objective function, making it suitable for complex, non-differentiable quantum optimization problems. Incorporating uncertainty from basis state measurements via the QM kernel enhances the algorithm's robustness, improving convergence and solution quality in noisy quantum environments.	Executing quantum circuit multiple times to estimate basis state distributions increases computational overhead, limiting scalability in resource-constrained quantum systems. The algorithm's performance can be sensitive to hyperparameters such as the scaling factor ω in the QM kernel, requiring careful tuning for optimal results, which may be challenging in practice.

IV. Key Metrics for Evaluating the Performance of the QAOA

Performance of QAOA is evaluated using different metrics, such as depth, approximation ratio, and number of qubits in quantum circuit.

A. Depth (p-level)

The p-level of QAOA breakdowns into the number of alternating layers of unitary operations, that are used. Specifically, each layer is composed of a problem Hamiltonian and a mixing Hamiltonian. The following are key points about the significance of depth:

1) Performance Improvement:: In general, higher number of p improves algorithms efficiency to attach near-optimal approximate solutions. It is right because the higher the number of p-levels, the more substantially changes in the optimization landscape can be captured.

2) Resource Demand:: On the other hand, the increased depth leads to a higher computational resource and longer execution time. The reason is that deeper circuits require more precise control and can cause more problem to noise and imperfection in current devices.

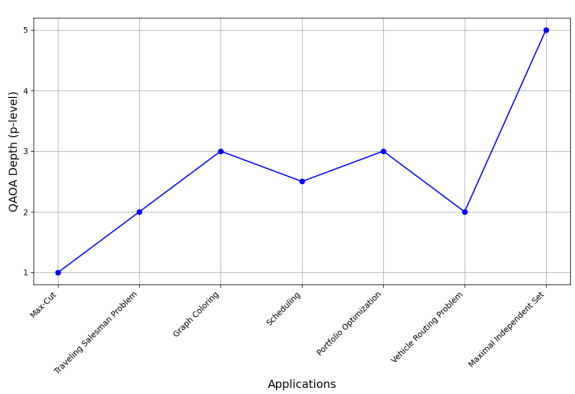

Fig. 3. QAOA Deapth (p-level) over various Applications

B. Approximate Ratio

Approximate ratio is an important metric of how well a solution Constructive for the Problems by QAOA. This ratio is calculated based on the value of a solution of QAOA and the value of any optimal solution.

1) Solution effectiveness: This ratio demonstrates how proximate is the QAOA solution to the optimal solution. A smaller approximation ratio demonstrates a solution closer to the optimal solution.

2) Algorithm efficiency: Calculating approximate ratio for different examples and p-levels for approximation help realize the effectiveness of using QAOA for solving specific programs.

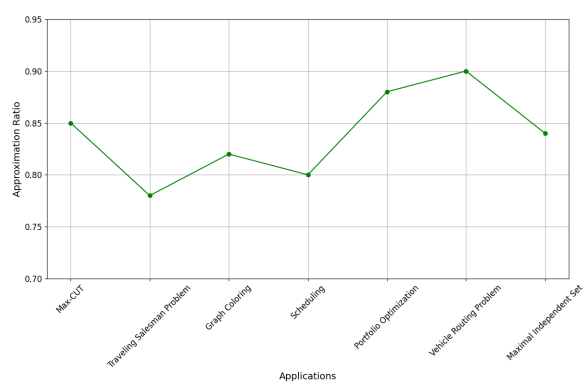

Fig. 4. QAOA Approximation Ratio over various Applications

C. Number of Qubits

The numbers of qubits used in a quantum circuit is directly related to the size of the problem that QAOA can be applying to. This is important for the following reasons:

1) Problem size: Increasing the size of the problem implies that we would be requiring more qubits. The number of qubits required with respect to the size of the problem is a crucial part of the consideration of how useful x would be when we transfer its application to the real-world scenarios.

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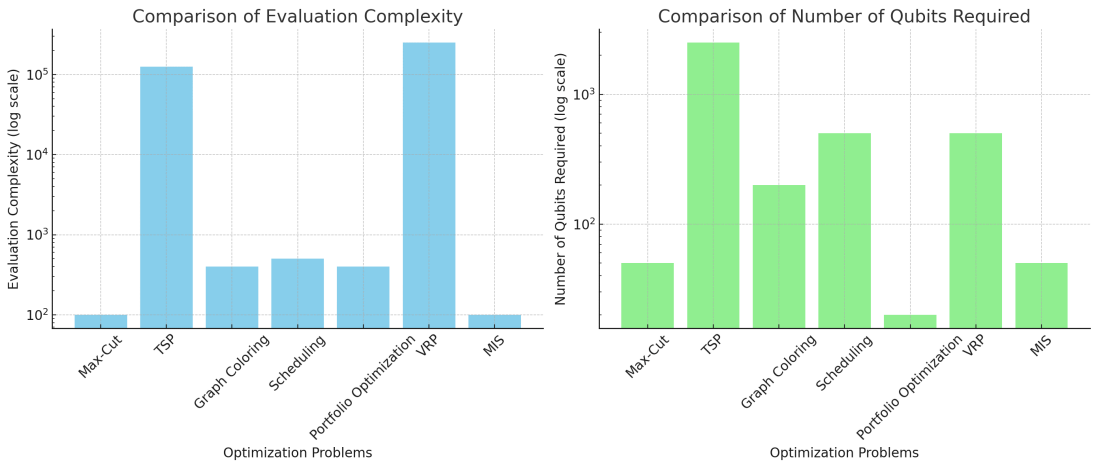

Fig. 5. Comparison of Evaluation Complexity and No: of Qubits Required for Different Optimization Problems

2) Quantum hardware constraints: The current state of quantum computing is at a stage where the practical management and control of increasing number of qubits is in a less manageable state. This implies that how effective or feasible QAOA is in its current form depends on how the requirements of qubits can be met efficiently on the hardware that is available.

V. CONCLUSION

Due to combinatorial nature of physical systems, QAOA might be very effective for np-hard problems such as the Vehicle Routing Problem. But practical concerns like inefficiencies when scaling and the need for large number of qubits and computational power make it difficult to apply in realistic situations. Deepening the QAOA over larger graph problems is also challenging for variational parameter optimization, which limits efficiency of both convergence and solution quality.
While these challenges are not trivial, some work is currently being done towards overcoming them. on the optimization side we need more effective optimization techniques and different circuit designs for scalability/robustness against noise to increase the efficiency of QAOA. Though QAOA faces challenges with larger problems, algorithmic and quantum hardware improvements could eventually make it a breakthrough innovation in quantum computing. This gives way to new approaches and strategies in Future Research, something that classical methods cannot offer thus showing the great potential of QAOA.

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